

Volume XVI: Scheme-Qualified Microscopic TMDs, Resolved Nuclear Evolution, and Process-Ready Accuracy Manifests

Audited matching data, common Collins–Soper evolution, rank-aware operator transport,
resolved spin-1 nuclear components, and qualified handoff to factorized observables

DeuteronWigner project

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Status and purpose. C19/M0 established a closed 540-dimensional regulated light-front and QCD operator basis, finite scheme transformations, regulator step scaling, rank-zero through rank-three transforms, analytic small- b_{TMD} matching, and controlled two-scale evolution oracles. Of the 540 basis entries, 492 are executable and 48 are explicitly unavailable; five shared matching parameters are constrained by eight conditions with three holdouts. These are architecture and validation results, not physical matching coefficients or physical evolution. C20/M1 is the source-audit package that replaces analytic coefficient oracles with primary-source perturbative records and compatible external or lattice constraints. The present volume defines the formal object that may emerge only after those source, scheme, covariance, and holdout gates pass: a scheme-qualified microscopic TMD ensemble at a reference scale, together with common rank-aware multi-scale evolution and process-readiness records. It does not assume that C20 has completed successfully.

Abstract

The microscopic program has reached a stage at which one regulated light-front Hamiltonian hierarchy produces quark, antiquark, and gluon GTMD helicity matrices; dynamical Wilson-line phases; a normalized spin-1 nuclear state containing nucleonic, pion, $\Delta\Delta$, and compact six-quark sectors; and an operator-level light-front-to-QCD matching skeleton. The remaining problem is not to add another named TMD ansatz. It is to qualify every executable operator in a single ultraviolet, rapidity, soft, Wilson-link, color, and Fourier scheme; to replace analytic coefficient oracles with source-audited perturbative and external matrix-element information; to evolve all transverse ranks and resolved nuclear components through one common (μ, ζ) system; and to expose the perturbative bottleneck and factorization status before any process calculation is authorized.

This volume defines the post-M1 object *scheme-qualified microscopic TMD ensemble*. The matching is a shared linear map between closed operator bases, with local-current anchors, regulator step scaling, compatible external or lattice matrix elements, source-provenance hashes, explicit missing-operator components, and a low-dimensional discrepancy basis where perturbation theory alone is insufficient. The Collins–Soper kernel is represented separately in the quark and gluon color representations, with perturbative small- b information, nonperturbative large- b constraints, and a covariance that is common across target polarization rather than refitted for each TMD. Rank-aware Fourier–Bessel transforms preserve the tensor harmonic, reference mass, and derivative-moment convention. Nonsinglet, singlet, helicity, transversity, gluon double-flip, and multiparton twist-three evolution remain distinct operator systems. The full nuclear graph is evolved without collapsing NN , $NN\pi$, $\Delta\Delta$, six-quark cluster, hidden-color, transition, and coherent components into one scalar response.

The formalism introduces source-audited coefficient records, scheme-qualified operator bundles, anomalous-dimension libraries, Collins–Soper kernel ensembles, evolution paths and curl manifests,

resolved nuclear evolution graphs, multi- Q member stores, process-readiness certificates, and authoritative accuracy tuples. It specifies staged implementation packages, analytic benchmarks, deliberately failing tests, and final gates for the handoff to qualified DY, SIDIS, inclusive b_1 , and gluon-sensitive process records. Passing these gates does not by itself establish a physical global extraction or a process prediction; it establishes that the complete microscopic state can be transported in one declared QCD scheme without erasing its flavor, spin, rank, Wilson-link, color, nuclear-sector, or uncertainty identity.

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1 Scope, inputs, and scientific boundary

1.1 Input chain

The input is the resolved microscopic operator vector at light-front regulator level r ,

$$\widetilde{\mathbf{W}}_{h,r}^{\text{LF}} = \left\{ \langle h', \Lambda' | \widetilde{\mathcal{O}}_{A,r}^{\text{LF}}(x, \mathbf{b}_{\text{TMD}}, \boldsymbol{\Delta}_T) | h, \Lambda \rangle \right\}_A, \quad (1)$$

where $h \in \{p, n, D\}$, and the deuteron state retains the resolved sector graph

$$\mathcal{G}_D = \{NN, NN\pi, \Delta\Delta, 6q_{\text{cluster}}, 6q_{\text{hidden}}, \text{transition/interference, coherent pilot}\}. \quad (2)$$

The label A contains species, flavor, target and parton polarization, transverse rank, Wilson order, link direction, ordered gluon-link pair, f/d color class, nuclear ancestry, and local-current or EMT moment status.

C19/M0 supplies two finite-related validation schemes and a closed basis with 540 entries on each side. It establishes that 492 entries are executable at the validation level and that 48 are fail-closed because the required operator or coefficient is absent. It also establishes a five-parameter shared matching skeleton with eight calibration conditions and three holdouts. These numbers are implementation facts, not evidence that the coefficients are physical.

1.2 Output chain

The intended output is

$$\widetilde{\mathbf{F}}_h^{\text{QCD}}(x, \mathbf{b}_{\text{TMD}}; \mu, \zeta, \mathfrak{s}) = \mathcal{U}_{\mathfrak{s}}[(\mu, \zeta) \leftarrow (\mu_0, \zeta_0)] \left[\mathbf{Z}_{\text{LF} \rightarrow \text{QCD}}^{\mathfrak{s}} \otimes_x \widetilde{\mathbf{W}}_{h,r}^{\text{LF}} + \boldsymbol{\Delta}_{\text{trunc}} \right], \quad (3)$$

with one scheme $\mathfrak{s}$, one correlated microscopic member, one resolved nuclear identity, and one accuracy manifest. A process object may consume equation (3) only after a separate factorization-readiness certificate supplies the hard, fragmentation, color, Glauber, measurement, and fixed-order information.

Design principle 1.1 (One operator map, many projections). *A matching or evolution parameter belongs to an operator block. It propagates to every TMD, GPD, PDF, current, EMT, Wigner, nuclear, and process projection derived from that block. It may not be attached directly to one failed named TMD.*

1.3 What this volume does not claim

This volume does not claim:

- that C20/M1 has passed its source-audit and external-constraint gates;
- that all 540 basis entries are physically matchable at one common perturbative order;
- that the 48 unavailable C19 entries are negligible or zero;
- that a lattice quasi-TMD, quasi-wave function, or Collins–Soper observable is already the same light-cone operator as the project TMD;
- that a four-loop rapidity anomalous dimension determines the nonperturbative large- b Collins–Soper kernel;
- that target polarization generates a separate universal evolution kernel;
- that a scheme-qualified TMD automatically has a valid process factorization theorem;
- that a successful matching validation constitutes a global extraction or a withheld prediction.

2 Evidence, provenance, and availability language

2.1 Evidence classes

Status	Meaning
EXACT	Algebraic identity, type rule, symmetry relation, or all-order consistency condition within the declared definitions.
REGULATOR_EXACT	Exact inside a fixed Hamiltonian basis, regulator, Fock truncation, or finite matrix representation.
PERTURBATIVE_SOURCE AUDITED	Coefficient or anomalous dimension transcribed from a primary source, with scheme, order, equation/table locator, source hash, and independent checks.
NONPERTURBATIVE INPUT	Common kernel or matrix element supplied by a microscopic, lattice, or phenomenological determination in a declared scheme.
MATCHED_MODEL	Shared low-dimensional matching or discrepancy operator constrained by several matrix elements and tested on holdouts.
CONTROLLED	Approximation with a declared expansion parameter, truncation order, and remainder or convergence test.
EXPLORATORY	A process, coefficient, or power correction used only for sensitivity studies.
UNAVAILABLE	A required ingredient is absent; authoritative composition must fail closed.

2.2 Primary-source coefficient record

Definition 2.1 (Coefficient provenance record). *A perturbative coefficient record is*

$$\mathbf{c} = (A \leftarrow B, a \leftarrow j, \text{twist}, m, \gamma, \mathcal{C}, \mathfrak{s}, p_{\text{first}}, p_{\text{impl}}, \mathcal{D}_x, \mathcal{S}_{\text{src}}, \mathcal{H}_{\text{src}}, \mathcal{O}_{\text{rem}}), \quad (4)$$

where $\mathcal{D}_x$ stores the distributional structure in x , $\mathcal{S}_{\text{src}}$ stores the primary citation and exact locator, $\mathcal{H}_{\text{src}}$ stores source and transcription hashes, and $\mathcal{O}_{\text{rem}}$ stores the known higher-order or power remainder.

A record becomes executable only after it passes:

1. tree or first-nonzero-order limits;
2. renormalization-group consistency;
3. Mellin moments and conserved sum rules where applicable;
4. endpoint and plus-distribution tests;
5. color-factor and crossing checks;
6. scheme covariance;
7. an independent analytic or numerical oracle.

Requirement 2.2 (No coefficient by analogy). *A coefficient for one polarization, transverse rank, Wilson-link class, or twist may not be assigned to another channel solely because the named functions have similar notation.*

2.3 External matrix-element provenance

A lattice, continuum, or experimental matrix-element bundle is identified by

$$\mathfrak{e} = (\mathcal{O}, |\chi_i\rangle, |\chi_j\rangle, \text{kinematics}, \mathcal{R}_{\text{ext}}, \mathfrak{s}_{\text{ext}}, \mathcal{Z}_{\text{ext} \rightarrow \text{QCD}}, C_{\text{cov}}, \\ \text{finite-volume status, continuum status, data ancestry, role}).$$

The role is one of CALIBRATION, DIAGNOSTIC, or HOLDOUT. Shared gauge ensembles, renormalization constants, momentum extrapolations, or experimental normalizations generate off-diagonal covariance and may not be duplicated as independent likelihood terms.

3 Scheme-qualified microscopic operator bundles

3.1 Complete scheme identity

The TMD scheme tuple is

$$\mathfrak{s}_{\text{TMD}} = (\mathcal{R}_{\text{UV}}, \mathcal{R}_{\text{rap}}, \mathcal{S}_{\text{soft}}, \text{square-root partition}, \gamma_{\text{staple}}, \text{cusp and closure}, \\ R_c, \mu, \zeta, n_f, \text{threshold history, Fourier convention}, \mathfrak{r}, \mathfrak{A}).$$

where $\mathfrak{r}$ is the transverse-rank identity and $\mathfrak{A}$ is the perturbative accuracy tuple.

Requirement 3.1 (Coordinate separation). *The Wigner coordinate $\mathbf{b}_\Delta$, conjugate to Δ_T , and the TMD impact parameter $\mathbf{b}_{\text{TMD}}$, conjugate to $\mathbf{k}_T$, remain distinct software and mathematical types. A matching or evolution map defined on $\mathbf{b}_{\text{TMD}}$ cannot consume $\mathbf{b}_\Delta$.*

3.2 Qualified operator bundle

Definition 3.2 (Scheme-qualified microscopic TMD bundle). *For one microscopic member s , define*

$$\mathfrak{F}_s = \left(\tilde{\mathbf{F}}_s^{\text{QCD}}(\mu_0, \zeta_0), \mathfrak{s}_{\text{TMD}}, \mathbf{Z}_s, \mathbf{C}_s, \Delta_s, \mathfrak{E}_s, \mathfrak{N}_s, \mathfrak{P}_s \right), \quad (5)$$

where $\mathbf{C}_s$ is the small- b coefficient basis, $\mathfrak{E}_s$ is the evolution manifest, $\mathfrak{N}_s$ is the resolved nuclear identity, and $\mathfrak{P}_s$ is the process-readiness status.

The bundle does not contain only central arrays. It carries matching covariance, coefficient-order metadata, missing-operator components, regulator comparison maps, numerical transform errors, and all unresolved statuses.

3.3 Availability lattice

Each operator entry has one of the following statuses:

Status	Interpretation
REFERENCE_SCALE_QUALIFIED	The LF-to-QCD map, UV/rapidity/soft identity, and coefficient record pass at the reference scale.
EVOLUTION_QUALIFIED	The entry has a compatible anomalous-dimension and Collins–Soper route over the requested scale domain.

Status	Interpretation
EXTERNAL_CONSTRAINED	A compatible external or lattice bundle contributes to the shared matching or kernel covariance.
PERTURBATIVE_WINDOW_ONLY	The entry is qualified only where $1/b$, external virtualities, and regulator scales lie in the declared perturbative window.
INDUCED_WITH_REMAINDER	A missing explicit sector is represented by a transformed induced operator and a nonzero remainder.
UNAVAILABLE_COEFFICIENT	The correct coefficient basis is absent.
UNAVAILABLE_EVOLUTION	Matching exists, but the appropriate collinear or rapidity evolution block is absent.
PROCESS_UNQUALIFIED	The TMD exists as an intermediate operator but cannot enter the requested factorization theorem.

4 Shared LF-to-QCD matching beyond analytic oracles

4.1 General map

The operator-level matching is

$$\tilde{\mathbf{F}}^{\text{QCD}} = \mathbf{Z}_{\text{LF} \rightarrow \text{QCD}} \otimes_x \tilde{\mathbf{W}}_r^{\text{LF}} + \tilde{\boldsymbol{\delta}}_{\text{trunc}}, \quad (6)$$

where $\mathbf{Z}$ may mix quark singlet, gluon, total-derivative, equation-of-motion, and multiparton operators when allowed by symmetry and twist.

For calibration states $|\chi_i\rangle$,

$$\mathbf{y}_{ij}^{\text{QCD}} = \mathbf{Z}(\theta_Z) \otimes_x \mathbf{y}_{ij}^{\text{LF}} + \mathbf{B} \theta_\Delta + \boldsymbol{\epsilon}_{ij}, \quad (7)$$

with shared matching parameters θ_Z , a named low-dimensional discrepancy basis $\mathbf{B}$, and covariance containing microscopic, perturbative, external, and numerical components.

Requirement 4.1 (Overconstrained matching). *The number and diversity of independent matrix elements must exceed the number of shared matching and discrepancy parameters. Jacobian rank, singular values, null directions, and holdout residuals are mandatory outputs.*

C19 demonstrates the architecture with five shared parameters, eight conditions, and three holdouts. A post-M1 result must retain at least this overconstraint and must not increase the executable count by adding one parameter per missing TMD.

4.2 Admissible information hierarchy

A matching plan may combine:

1. perturbative-window coefficients;
2. regulator step scaling;
3. exact or renormalized local-current and EMT anchors;
4. compatible external or lattice matrix elements;
5. a small shared discrepancy basis;

6. explicit missing-operator remainders.

The plan is fixed before holdout evaluation. One channel may not use a perturbative route and another an external route merely because the latter gives a preferred result, unless the difference is part of the declared operator plan.

4.3 Finite scheme transformations

For an invertible finite transformation R ,

$$\tilde{F}^{s'} = R^{s' \leftarrow s} \otimes_x \tilde{F}^s. \quad (8)$$

Any hard factor or later asymptotic subtraction must transform covariantly. A scheme round trip is an exact algebraic benchmark; its numerical residual may not be included as physical uncertainty.

5 Regulator step scaling and the continuum trajectory

For regulator levels $r \preceq r'$, define

$$\Sigma_{r' \leftarrow r} = Z_{\text{LF} \rightarrow \text{QCD}}(r')^{-1} Z_{\text{LF} \rightarrow \text{QCD}}(r). \quad (9)$$

The cocycle defect is

$$\delta_{r''r'r} = \Sigma_{r'' \leftarrow r'} \Sigma_{r' \leftarrow r} - \Sigma_{r'' \leftarrow r}. \quad (10)$$

C19 closes this defect for its analytic validation system. C20 and later packages must decompose any nonzero physical-source defect into:

$$\delta = \delta_{\text{pert}} + \delta_{\text{basis/Fock}} + \delta_{\text{missing op}} + \delta_{\text{external}} + \delta_{\text{scheme}} + \delta_{\text{num}} + \delta_{\text{disc}}. \quad (11)$$

Proposition 5.1 (Observable-level comparison). *Bare Fock probabilities and basis coefficients at different regulator levels need not converge to equal values. The convergence target is a set of matched matrix elements or observables transported to one common scheme.*

6 Reference-scale boundary and uncertainty ownership

At (μ_0, ζ_0) , the qualified boundary is

$$\tilde{F}_{s,0} = Z_s \otimes_x \tilde{W}_{s,r}^{\text{LF}} + \Delta_{s,\text{trunc}}. \quad (12)$$

The covariance is partitioned as

$$\begin{aligned} C_0 = & C_{\text{micro}} + C_{\text{Hamiltonian/Fock}} + C_{\text{regulator}} + C_{\text{coefficient}} + C_{\text{matching}} + C_{\text{external}} \\ & + C_{\text{soft/rapidity}} + C_{\text{nuclear}} + C_{\text{transform}} + C_{\text{numerical}}. \end{aligned} \quad (13)$$

These components are retained separately before any optional combined summary.

Requirement 6.1 (Member integrity). *One posterior or validation member carries its proton, neutron, deuteron, Wilson, matching, Collins–Soper, nuclear, and numerical identities through every scale. Marginal bands from unrelated members may not be combined as though they were a correlated ensemble.*

7 Physical anomalous-dimension library

7.1 Library identity

An anomalous-dimension set is

$$\mathfrak{d}_a = \left(\Gamma_{\text{cusp}}^a, \gamma_F^a, \mathcal{D}_a^{\text{pert}}, \beta, p_\Gamma, p_\gamma, p_{\mathcal{D}}, p_\beta, \mathfrak{s}, n_f, \text{sources and hashes} \right). \quad (14)$$

The library must include convention adapters for factors of two, signs, $\ln \zeta$ versus $\ln \sqrt{\zeta}$, and the definition of the rapidity anomalous dimension.

The rapidity anomalous dimension is known perturbatively through four loops in QCD [13, 14]. This does not imply that every matching coefficient, hard factor, polarized splitting function, or process subtraction is known at the same order.

7.2 Source audit and algebraic checks

Every library member must pass:

- cusp and noncusp convention conversions;
- the mixed-derivative consistency relation;
- expansion against lower-order known results;
- representation and n_f checks;
- threshold compatibility;
- a source-level independent implementation check.

Requirement 7.1 (No global accuracy from one ingredient). *The highest known order of Γ_{cusp} or the rapidity anomalous dimension does not determine the accuracy of an observable. The bottleneck among matching, hard, collinear, rapidity, fixed-order, and process ingredients is authoritative.*

8 Collins–Soper kernel across the full transverse domain

8.1 Representation-level kernel

For parton representation $a = q, g$, write

$$\mathcal{D}_a(b; \mu) = \mathcal{D}_a^{\text{pert}}(b_*; \mu) + \mathcal{D}_a^{\text{NP}}(b) + \delta \mathcal{D}_a^{\text{match}}(b). \quad (15)$$

The kernel is common across target polarization, flavor within the same representation, and nuclear sector at leading power. Polarization and nuclear dependence reside in the boundary and matching coefficients.

8.2 Small- b perturbative domain

The perturbative kernel uses the source-audited rapidity anomalous dimension and a declared profile or matching point. The profile identity contains b_* , b_{max} , μ_b , scale caps, and transition derivatives. Profile variation is distinct from nonperturbative-kernel uncertainty.

8.3 Large- b nonperturbative domain

The nonperturbative kernel may be constrained by:

- global DY, vector-boson, and SIDIS analyses;
- lattice quasi-TMD or TMD-wave-function determinations after controlled matching;
- energy-energy-correlator or other universal-kernel extractions;
- the microscopic model only if it includes the same soft operator and avoids overlap with the Wilson phase budget.

Recent lattice calculations determine the quark Collins–Soper kernel with controlled continuum, mass, and matching studies [18, 20]. These results are external constraints in a declared scheme. They are not silently copied into the gluon kernel or into a target-polarization-dependent kernel.

Requirement 8.1 (Kernel universality challenge). *A common quark or gluon kernel must describe all supported target and polarization channels after their distinct boundaries and coefficient functions are included. A failed channel first challenges matching, power corrections, or the boundary; it does not immediately receive its own Collins–Soper kernel.*

9 Two-scale evolution as a geometric transport

9.1 Evolution equations

Use the convention

$$\frac{d}{d \ln \mu} \ln \tilde{F}_a(x, b; \mu, \zeta) = \gamma_F^a(\mu, \zeta), \quad (16)$$

$$\frac{d}{d \ln \sqrt{\zeta}} \ln \tilde{F}_a(x, b; \mu, \zeta) = -\mathcal{D}_a(b; \mu), \quad (17)$$

with

$$\frac{\partial \gamma_F^a}{\partial \ln \sqrt{\zeta}} = -\Gamma_a, \quad \frac{d \mathcal{D}_a}{d \ln \mu} = \Gamma_a. \quad (18)$$

The evolution one-form is

$$\omega_a = \gamma_F^a d \ln \mu - \mathcal{D}_a d \ln \sqrt{\zeta}. \quad (19)$$

At all orders, $d\omega_a = 0$. At finite order, define the curl defect

$$\Omega_a^{(N)} = \frac{\partial \gamma_F^{a,(N)}}{\partial \ln \sqrt{\zeta}} + \frac{d \mathcal{D}_a^{(N)}}{d \ln \mu}. \quad (20)$$

C19 deliberately retains a finite-order curl of 0.0037 and path difference 0.0019 in its analytic truncated oracle. A physical implementation reports and propagates this defect rather than refitting it away.

9.2 Admissible paths

The supported path classes are:

1. direct rectangular or piecewise contours;
2. integrability-improved evolution;

3. a declared ζ -prescription or null-evolution curve;
4. profile-based contours with explicit transition derivatives.

The path identity and finite-order restoration rule are part of the result. Two paths may be compared only when they use the same anomalous-dimension, kernel, threshold, and scheme data.

9.3 Evolution transitivity

For scales A, B, C ,

$$\mathcal{U}(C \leftarrow B)\mathcal{U}(B \leftarrow A) = \mathcal{U}(C \leftarrow A) + \delta_{CBA}^{\text{evol}}. \quad (21)$$

The defect is reported separately from scale variation and from the LF-to-QCD matching uncertainty.

10 Rank-aware Fourier–Bessel transport

For transverse harmonic m ,

$$\tilde{F}^{(m)}(x, b) = 2\pi i^m \int_0^\infty k \, dk \, J_{|m|}(bk) N_m(k) F^{(m)}(x, k), \quad (22)$$

with inverse

$$F^{(m)}(x, k) = \int_0^\infty \frac{b \, db}{2\pi} (-i)^m J_{|m|}(bk) \tilde{N}_m(b) \tilde{F}^{(m)}(x, b). \quad (23)$$

The registry stores m , Bessel order, Fourier phase, extracted k/M and bM powers, reference mass, and derivative-moment convention.

Proposition 10.1 (Rank preservation). *At leading power, radial (μ, ζ) evolution does not change the transverse $SO(2)$ weight. Any apparent rank mixing signals an operator-basis, transform, or power-expansion issue.*

C19 closes rank-zero through rank-three analytic transforms within 8×10^{-8} . A post-M1 implementation must retain this transform residual separately from matching and evolution errors.

10.1 Derivative moments

Weighted transverse moments are defined through declared derivatives in b -space. For example,

$$F^{(1)}(x) = \int d^2 \mathbf{k}_T \frac{\mathbf{k}_T^2}{2M^2} F(x, \mathbf{k}_T^2) \longleftrightarrow -\frac{2}{M^2} \frac{\partial}{\partial b^2} \tilde{F}(x, b) \Big|_{b=0}, \quad (24)$$

only after the ultraviolet and operator-renormalization status of the moment is declared.

11 Source-audited small- b OPE

11.1 General expansion

$$\tilde{F}_A^{a/h}(x, b; \mu, \zeta) = \sum_{B,j} C_{A \leftarrow B}^{a \leftarrow j}(x, b; \mu, \zeta) \otimes_x f_B^{j/h}(x; \mu) + \mathcal{O}((b\Lambda)^p). \quad (25)$$

The coefficient identity includes source and target operators, species, target channel, transverse rank, twist, link/color class, first nonzero order, implemented order, and power remainder.

11.2 Twist-two coefficient families

The source-audited library may support, at their actual available orders:

- unpolarized quark, antiquark, and gluon channels;
- quark and gluon helicity;
- quark transversity;
- linearly polarized gluons;
- rank-zero spin-1 tensor- LL matrix elements with the same local twist-two operator class;
- singlet quark–gluon mixing;
- local current and EMT moments.

High-order twist-two matching for linearly polarized gluons and quark transversity is available in the recent literature [15, 16]. Its existence does not upgrade unsupported spin-1 tensor, twist-three, hard-process, or nuclear coefficients.

11.3 Spin-1 tensor channels

For a rank-zero tensor-polarized matrix element of the same twist-two light-ray operator, target polarization changes the hadronic matrix element, not the short-distance coefficient. This statement must still be checked against the exact operator convention, normalization, and mixing block. Ranked tensor, double-helicity-flip, or genuinely many-body nuclear operators may require distinct coefficient bases.

11.4 Twist-three and link-odd channels

The following remain separate multiparton bases:

$$\left\{ T_F^q, T_F^{(\sigma)q}, T_G^{(f)}, T_G^{(d)}, \text{genuine worm-gear operators, tensor-polarized multiparton operators} \right\}. \quad (26)$$

A twist-two coefficient cannot be substituted for Sivvers, Boer–Mulders, or gluonic-pole matching. If the correct source-audited coefficient is absent, the entry remains unavailable.

11.5 Power domain and overlap

The OPE carries a declared validity domain in b , x , scale, and regulator. The profile joining perturbative and nonperturbative domains must not count the same soft or power contribution in both the microscopic boundary and the Collins–Soper kernel.

12 Collinear evolution and thresholds

12.1 Nonsinglet and singlet systems

Nonsinglet evolution is

$$\frac{dq_{\text{NS}}}{d \ln \mu^2} = P_{\text{NS}} \otimes_x q_{\text{NS}}. \quad (27)$$

The singlet system is

$$\frac{d}{d \ln \mu^2} \begin{pmatrix} \Sigma \\ g \end{pmatrix} = \begin{pmatrix} P_{qq} & P_{qg} \\ P_{gq} & P_{gg} \end{pmatrix} \otimes_x \begin{pmatrix} \Sigma \\ g \end{pmatrix}. \quad (28)$$

The tensor-polarized rank-zero singlet pair $(\delta_T \Sigma, \delta_T g)$ uses the corresponding twist-two local operator mixing when the operator basis and convention match.

12.2 Polarized and exotic sectors

Separate systems are retained for:

- helicity singlet and nonsinglet evolution;
- quark transversity;
- gluon double-helicity-flip operators;
- multiparton twist-three quark–gluon and tri-gluon operators;
- pion, transition, coherent, and compact nuclear operators when they have independent scale dependence.

Recent high-order polarized matching results also provide corresponding splitting-function information, but the implemented accuracy must be recorded channel by channel [17].

12.3 Heavy-flavor thresholds

At $\mu = m_h$,

$$\mathbf{f}^{(n_f+1)}(m_h) = \mathbf{A}_h(m_h) \otimes_x \mathbf{f}^{(n_f)}(m_h), \quad (29)$$

with simultaneous matching of α_s , collinear operators, TMD coefficients, anomalous dimensions, and hard factors. A change of n_f without the threshold map fails closed.

13 Resolved nuclear evolution

13.1 Operator graph remains resolved

The deuteron bundle retains

$$\mathfrak{N}_D = \{NN, NN\pi, \Delta\Delta, 6q_{\text{cluster}}, 6q_{\text{hidden}}, \text{transition/interference, coherent pilot, matched total}\}. \quad (30)$$

Evolution is applied to operator blocks, not to labels such as “nuclear correction.”

13.2 Impulse commutation

For a scale-independent impulse kernel S ,

$$P \otimes_x (S \otimes_x f) = S \otimes_x (P \otimes_x f). \quad (31)$$

C19 closes this analytic associativity to 2.7×10^{-13} . The identity fails when the nuclear kernel has independent scale dependence or when two-body, pion, coherent, or compact operators mix under renormalization. Those blocks require their own matching and evolution maps.

13.3 Pion and transition operators

The pion-active parent remains unmatched until a compatible pion operator and scheme are supplied. Continuum calibration of the $NN\pi$ transition does not create a physical pion TMD. Transition and exchange-current operators may mix with contact operators and must preserve the C17 separator-flow relation.

13.4 $\Delta\Delta$, six-quark, and hidden color

The $\Delta\Delta$ and compact sectors receive explicit operator statuses. The hidden-color basis may rotate by $U(4)$, but complete six-quark and matched observables must remain invariant. C19 closes this covariance to 1.5×10^{-13} in its analytic matching skeleton. A post-M1 coefficient or evolution block that depends on a hidden-color basis vector rather than the complete operator subspace fails.

13.5 Coherent amplitudes

A coherent small- x amplitude is not evolved as an unpolarized scalar ratio. Its helicity-resolved diffractive operator, propagation kernel, and parton–nuclear overlap subtraction carry independent scale and factorization statuses. Until physical diffractive inputs and a qualified factorization map exist, this branch remains exploratory.

14 Common multi- Q ensemble and covariance

For member s , define

$$\mathcal{E}_s(Q) = \left\{ \tilde{F}_{A,s}^{a/h}(x, b; Q), \tilde{F}_{A,s}^{a/D}(x, b; Q), \mathfrak{A}_{A,s}, \mathfrak{N}_s, \mathfrak{s}_s, \mathfrak{U}_s, C_s \right\}. \quad (32)$$

The same member is evaluated at all requested scales. Cross-scale covariance is

$$\text{Cov}[F_i(Q_1), F_j(Q_2)] = \left\langle \left(F_i^{(s)}(Q_1) - \bar{F}_i(Q_1) \right) \left(F_j^{(s)}(Q_2) - \bar{F}_j(Q_2) \right) \right\rangle_s. \quad (33)$$

The ensemble retains cross-flavor, quark–gluon, proton–neutron, nucleon–deuteron, rank, Wilson-link, color, nuclear-sector, matching, and scale correlations.

15 Process-readiness certificates

A scheme-qualified TMD is not automatically a process prediction. Define

$$\mathfrak{P}_{\text{ready}} = (\text{process, external states, hard channel, TMD/FF basis,} \\ \text{link map, color map, soft prescription, measurement,} \\ \text{power counting, Glauber status, factorization status, fixed-order status}).$$

The status is one of:

Status	Meaning
ESTABLISHED	The required factorization, link, color, soft, and power-counting structure is established for the declared observable.
CONDITIONAL	A theorem or cancellation holds under explicit hypotheses that are carried in the record.
EXPLORATORY	A sensitivity calculation is permitted, but not an authoritative prediction.
BROKEN	Known color entanglement, Glauber exchange, or another obstruction invalidates the standard separate-hadron TMD composition.
UNASSESSED	The process cannot consume the TMD bundle.

15.1 Initial process families

The first process-ready records after M2 should be:

- color-singlet DY, including tensor-polarized deuteron combinations;
- SIDIS with compatible TMD fragmentation functions and spin-1 projectors;
- inclusive b_1 and local-current closure as collinear validation, not TMD factorization;
- gluon-sensitive channels only after the ordered-link, f/d color, hard, and Glauber status are supplied.

No default process-level $f + d$ gluon mixture is permitted.

16 Process-ready accuracy manifests

16.1 Authoritative accuracy tuple

The accuracy is

$$\mathfrak{A} = (p_\Gamma, p_\gamma, p_{\mathcal{D}}, p_C, p_P, p_H, p_{FF}, p_{FO}, p_{\text{thr}}, p_{\text{nuc}}, p_{\text{Wilson}}). \quad (34)$$

A human-readable label such as N^kLL is generated only after the tuple is checked against the project convention.

16.2 Bottleneck rule

For an observable requiring several structures,

$$p_{\text{observable}} = \min \{p_\Gamma, p_\gamma, p_{\mathcal{D}}, p_C, p_P, p_H, p_{FF}, p_{FO}, p_{\text{thr}}\}. \quad (35)$$

Unknown process color weights, twist-three coefficients, or fixed-order tails cannot be hidden behind a high-order cusp result.

16.3 Uncertainty axes

The manifest reports separately:

1. microscopic parameter or posterior uncertainty;

2. Hamiltonian, Fock, basis, and regulator truncation;
3. Wilson-order and spectral-cut truncation;
4. LF-to-QCD matching and missing-operator uncertainty;
5. perturbative coefficient and anomalous-dimension truncation;
6. evolution path, curl, and scale variation;
7. nonperturbative Collins–Soper-kernel uncertainty;
8. collinear multiparton and threshold evolution uncertainty;
9. nuclear many-body, separator, coherent, and compact-sector truncation;
10. process-factorization and Glauber status;
11. fixed-order and later Y -term matching;
12. numerical transform, grid, interpolation, and quadrature error.

Convergence in one axis may not conceal nonconvergence in another.

17 Software architecture

Object	Responsibility
<code>CoefficientSourceRecord</code>	Primary source, equation/table locator, scheme, order, source hash, transcription hash, distributional convention, independent oracle, and remainder.
<code>ExternalMatrix</code>	Operator, state, scheme, kinematics, covariance,
<code>ElementBundle</code>	finite-volume/continuum status, ancestry, and calibration role.
<code>MatchingPlan</code>	Exclusive perturbative, external, hybrid, or validation route with parameter ownership and holdouts.
<code>MatchingPosterior</code>	Shared parameters, covariance, Jacobian rank, singular values, null
<code>OrFit</code>	directions, residual decomposition, and predictive holdouts.
<code>SchemeQualified</code>	Reference-scale QCD operator vector with complete scheme, matching,
<code>TMDParent</code>	missing-operator, nuclear, and member identity.
<code>AnomalousDimension</code>	Cusp, noncusp, rapidity, beta-function, threshold, convention, order,
<code>Library</code>	and source provenance.
<code>CollinsSoperKernel</code>	Quark/gluon perturbative and nonperturbative kernel members,
<code>Ensemble</code>	matching profile, external constraints, and covariance.
<code>EvolutionPath</code>	Endpoints, contour, profile, integrability restoration, curl, transitivity,
	solver, and threshold history.
<code>RankTransform</code>	Bessel order, phases, mass powers, quadrature, derivative moments,
	and round-trip reports.
<code>SmallBOPELibrary</code>	Source-audited operator coefficient matrix, twist, rank, link/color, first
	nonzero order, power domain, and availability.
<code>ResolvedNuclear</code>	Separate NN, pion, transition, $\Delta\Delta$, compact, hidden-color, coherent,
<code>EvolutionGraph</code>	and total blocks with matching/evolution maps.
<code>MultiQMemberStore</code>	Correlated microscopic member at all scales with matching, kernel,
	nuclear, and numerical identities.
<code>ProcessReadiness</code>	Hard, FF, link, color, soft, measurement, power, Glauber, factorization,
<code>Certificate</code>	and fixed-order statuses.

Object	Responsibility
AccuracyManifest	Full order tuple, bottleneck, scale plan, uncertainty axes, source hashes, and human label.
QualifiedClosure Report	Scheme, moment, rank, path, threshold, nuclear, hidden-color, link-reversal, and holdout residuals.

17.1 Fail-closed comparisons

Every map compares:

- b_{Δ} versus b_{TMD} ;
- rank, Bessel order, Fourier phase, and reference mass;
- quark, antiquark, fundamental, anti-fundamental, and adjoint identities;
- target and parton operator polarization;
- future/past and ordered gluon-link identity;
- f/d color class;
- UV, rapidity, soft, and finite scheme;
- μ, ζ, n_f , and threshold history;
- microscopic and nuclear member identity;
- matching and perturbative accuracy;
- process and Glauber status.

Array-shape compatibility is never sufficient.

18 Validation ladder and benchmark suite

18.1 Layered validation

The validation order is:

1. source and transcription provenance;
2. coefficient algebra and distributional limits;
3. operator-basis completeness and scheme covariance;
4. overconstrained matching and holdouts;
5. regulator step scaling;
6. rank-transform round trips;
7. exact and finite-order evolution geometry;
8. collinear moments and thresholds;
9. resolved nuclear commutation and hidden-color covariance;
10. multi-scale member and link-reversal preservation;

11. process-readiness rejection and acceptance tests.

18.2 Required benchmark families

ID	Benchmark
S-A	Primary-source coefficient transcription with an independent distributional oracle.
S-B	Finite scheme transformation acting covariantly on the operator vector and a mock hard factor.
S-C	Shared matching fit with known parameters, covariance, null-direction injection, and genuine holdouts.
S-D	Three-resolution step-scaling cocycle with one deliberately omitted induced operator.
S-E	Compatible and incompatible external/lattice bundle ingestion, including shared covariance.
S-F	Rank 0, 1, 2, 3 transform round trips and deliberate wrong-Bessel failures.
S-G	Exact two-scale potential with path independence and a truncated nonzero-curl oracle.
S-H	Quark and gluon Collins–Soper kernels with common polarization dependence and rejected polarization-specific aliases.
S-I	Nonsinglet and singlet evolution with conserved moments.
S-J	Heavy-flavor threshold continuity and reverse-map test.
S-K	Future/past link-reversal preservation through matching and evolution.
S-L	Resolved nuclear impulse commutation and a scale-dependent two-body counterexample.
S-M	Hidden-color basis rotation with invariant complete matched observables.
S-N	Multi- Q member covariance and transitivity.
S-O	Process-readiness certificate accepting qualified color-singlet channels and rejecting an unassessed gluon channel.
S-P	Accuracy-manifest bottleneck test in which a high-order cusp cannot upgrade a lower-order coefficient or fixed-order ingredient.

18.3 Deliberate negative tests

The implementation derived from this volume must include ordered failures for:

- missing or altered source hashes and equation locators;
- wrong plus-distribution, endpoint, or $\delta(1 - x)$ terms;
- coefficient assignment by TMD name rather than operator identity;
- twist-two coefficients used for twist-three channels;
- one normalization or width per named TMD;
- underconstrained matching and hidden null directions;
- incompatible lattice scheme, momentum, or finite-volume status;

- duplicated external-data ancestry;
- step-scaling cocycle violation;
- UV, rapidity, soft, or finite-scheme double counting;
- wrong-rank Fourier–Bessel transforms;
- polarization-specific Collins–Soper kernels without evidence;
- evolution curl or transitivity failures;
- threshold change without matching;
- collapsed nuclear sectors or hidden-color basis dependence;
- process use without a factorization/Glauber certificate;
- accuracy laundering;
- leakage into the accepted 216-route production model.

19 Staged implementation program

19.1 M1: source-audited matching completion

C20/M1 is responsible for:

1. the source-audited perturbative coefficient library;
2. compatible external/lattice matrix-element interfaces;
3. shared overconstrained matching plans;
4. physical-source step scaling and holdouts;
5. audited small- b coefficient statuses.

Its output remains a reference-scale validation bundle unless all scheme and source gates pass.

19.2 M2: physical evolution library and multi- Q ensembles

The expected next package is:

1. source-audited anomalous-dimension and threshold libraries;
2. quark and gluon Collins–Soper kernel ensembles;
3. compatible lattice or phenomenological kernel constraints;
4. exact, contour, integrability-improved, and ζ -prescription routes;
5. finite-order curl and transitivity covariance;
6. common multi- Q nucleon and deuteron TMD ensembles.

19.3 M3: full rank-aware OPE and collinear mixing

Implement:

1. audited twist-two coefficients at their real perturbative orders;
2. the correct twist-three and tri-gluon bases;
3. nonsinglet, singlet, helicity, transversity, double-flip, and multiparton evolution;
4. rank-aware derivative moments;
5. heavy-flavor threshold maps;
6. explicit unavailable statuses for incomplete channels.

19.4 M4: resolved nuclear evolution

Apply the common scheme to the complete nuclear graph, test impulse commutation where valid, and implement independent matching/evolution blocks for pion, transition, coherent, $\Delta\Delta$, and compact operators. Preserve hidden-color covariance and the C17 separator flow.

19.5 P0: process-readiness records

Before computing a cross section, implement qualified records for DY, SIDIS, inclusive b_1 , and selected gluon-sensitive channels. This package defines no $W+Y$ result until the same-scheme hard and fixed-order ingredients are available.

20 Risks, failure modes, and explicit nonclaims

Risk	Failure mode	Required safeguard
Matching by name	One finite factor per TMD.	Match closed operator blocks and overconstrain shared parameters.
Source drift	Coefficient code no longer corresponds to the cited equation/version.	Source, version, equation, and transcription hashes in every record.
Scheme drift	Soft, rapidity, Fourier, or hard conventions from different schemes are mixed.	Complete scheme tuple and finite adapters.
External-input aliasing	A quasi-TMD or CS observable is treated as the light-cone TMD itself.	Typed external matching formula and covariance.
Kernel overfitting	One Collins–Soper kernel is fitted per polarization.	Representation-level common kernel and universality challenge.
Rank erasure	Scalar transforms are applied to ranked operators.	Rank-aware transform identity and reconstruction tests.
Twist laundering	Twist-two matching is used for a twist-three T-odd function.	Explicit multiparton basis and fail-closed coefficient status.

Risk	Failure mode	Required safeguard
Nuclear collapse	All many-body mechanisms are evolved as one scalar correction.	Resolved nuclear operator graph.
Hidden-color artifact	Complete results change under a hidden-color basis rotation.	Covariant subspace matching and invariant totals.
Accuracy laundering	A high-order anomalous dimension labels a lower-order observable.	Full accuracy tuple and bottleneck rule.
Process overreach	A qualified TMD is used in a broken or unassessed process.	Process-readiness certificate and Glauber gate.
Patchwork return	Failed channels receive independent widths or normalizations.	Parameter ownership and withheld holdouts.

This volume does not claim a physical global TMD extraction, all-order evolution, complete T-odd matching, physical process predictions, or production readiness. It defines the formal conditions under which those claims may later become meaningful.

21 Formal acceptance criteria and handoff

A scheme-qualified multi-scale microscopic TMD package is accepted only when:

1. every executable coefficient has primary-source provenance and an independent check;
2. the LF and QCD bases are closed over the declared supported operator set;
3. unavailable entries remain explicit and fail closed;
4. shared matching parameters are overconstrained and genuine holdouts are reported;
5. UV, rapidity, soft, finite-scheme, LF-to-QCD, and truncation pieces remain distinct;
6. regulator step scaling closes within a decomposed uncertainty;
7. local currents, Ward identities, and EMT moments remain consistent after matching;
8. rank-zero through rank-three transforms close with the declared phases and powers;
9. the small- b OPE uses the correct twist, link, color, and coefficient order;
10. quark and gluon Collins–Soper kernels remain separate representation-level objects;
11. finite-order evolution curl and transitivity defects are reported and propagated;
12. nonsinglet, singlet, helicity, transversity, double-flip, and twist-three evolution remain distinct;
13. heavy-flavor thresholds preserve physical moments;
14. the resolved nuclear graph is not collapsed into a scalar correction;
15. complete results remain invariant under hidden-color basis rotations;
16. partonic link reversal is preserved at all scales;
17. one correlated microscopic member survives through every scale and target;

18. the accuracy manifest reports the true bottleneck;
19. process-unqualified objects cannot enter a cross section;
20. accepted production artifacts and the 216-route phenomenological registry remain unchanged until a separately authorized migration.

Passing these conditions establishes a scheme-qualified, process-ready intermediate operator ensemble. It does not by itself establish a process prediction, a global posterior, or a successful withheld observable.

The next formal handoff is to qualified process records and fixed-order matching. The next inference handoff occurs only after at least one multi-scale nucleon and deuteron observable family has been reserved as a genuine holdout.

A Minimal coefficient-source schema

```
{
  "coefficient_id": "...",
  "source_operator": "...",
  "target_operator": "...",
  "species_block": "...",
  "twist": 2,
  "transverse_rank": 0,
  "wilson_link_class": "...",
  "gluon_color_class": "NOT_APPLICABLE|F_TYPE|D_TYPE",
  "scheme_id": "...",
  "first_nonzero_order": "...",
  "implemented_order": "...",
  "distributional_convention": "...",
  "primary_source": "...",
  "equation_or_table": "...",
  "source_sha256": "...",
  "transcription_sha256": "...",
  "independent_oracle": "...",
  "power_remainder": "...",
  "availability": "..."
}
```

B Minimal scheme-qualified member schema

```
{
  "member_id": "...",
  "microscopic_state_id": "...",
  "nuclear_member_id": "...",
  "matching_plan_id": "...",
  "tmd_scheme_id": "...",
  "reference_scales": {"mu": "...", "zeta": "..."},
}
```

```

"coefficient_library_hash": "...",
"anomalous_dimension_library_hash": "...",
"cs_kernel_member_id": "...",
"evolution_path_id": "...",
"threshold_history": [...],
"rank_transform_manifest": "...",
"resolved_nuclear_graph": "...",
"matching_covariance": "...",
"evolution_covariance": "...",
"unavailable_entries": [...],
"process_readiness": [...]
}

```

C Minimal process-readiness schema

```

{
  "process_id": "...",
  "factorization_status": "ESTABLISHED|CONDITIONAL|EXPLORATORY|BROKEN|UNASSESSED",
  "external_states": [...],
  "hard_channel": "...",
  "tmd_and_ff_basis": [...],
  "link_map": "...",
  "gluon_color_map": "...",
  "soft_prescription": "...",
  "measurement_function": "...",
  "glauber_status": "...",
  "fixed_order_status": "...",
  "accuracy_manifest": "...",
  "missing_ingredients": [...]
}

```

D Summary of stable formal requirements

ID	Requirement
V16.SRC.1	Every executable perturbative coefficient has primary-source, locator, source-hash, transcription-hash, and independent-oracle records.
V16.SRC.2	External/lattice bundles carry compatible scheme conversion, covariance, continuum, and ancestry status.
V16.TYPE.1	Keep $\mathbf{b}_\Delta$, $\mathbf{b}_{\text{TMD}}$, $\mathbf{q}_T$, and loop momenta as distinct types.
V16.TYPE.2	Rank, Bessel order, Fourier phase, extracted powers, and reference mass are operator identity.
V16.SCHEME.1	Store one complete UV/rapidity/soft/link/color/Fourier/threshold scheme tuple.
V16.SCHEME.2	Finite transformations act covariantly on all later process ingredients.
V16.MATCH.1	Match closed operator bases, never named TMD curves.
V16.MATCH.2	Overconstrain shared parameters and report null directions and holdouts.
V16.MATCH.3	Preserve currents, Ward identities, EMT moments, and missing-operator remainders.
V16.STEP.1	Demonstrate regulator step scaling and decompose cocycle defects.
V16.BOUND.1	Export a reference-scale scheme-qualified operator vector with separated covariance axes.

ID	Requirement
V16.AD.1	Anomalous-dimension libraries carry source, convention, order, n_f , and threshold identity.
V16.CS.1	Keep quark and gluon Collins–Soper kernels separate and common across polarization.
V16.CS.2	Treat lattice or phenomenological kernel information as scheme-qualified external input, not a universal copy.
V16.EVOL.1	Use convention-consistent (μ, ζ) equations and report finite-order curl.
V16.EVOL.2	Report evolution path, transitivity, threshold history, and restoration prescription.
V16.RANK.1	Use $J_{ m }$ with declared phases and powers for every rank.
V16.OPE.1	Identify coefficient twist, rank, link/color class, order, and power domain.
V16.OPE.2	Keep twist-two and multiparton twist-three bases distinct.
V16.COLL.1	Separate nonsinglet, singlet, helicity, transversity, double-flip, and twist-three evolution.
V16.THR.1	Use explicit heavy-flavor threshold maps and conserve physical moments.
V16.NUC.1	Evolve the resolved nuclear operator graph; do not hide scale dependence in one scalar response.
V16.NUC.2	Preserve cluster subtraction, separator flow, and hidden-color basis covariance.
V16.ENS.1	Preserve one microscopic member across flavor, target, nuclear sector, rank, and scale.
V16.PROC.1	Require a factorization/Glauber/process-readiness certificate before cross-section assembly.
V16.ACC.1	Use a full accuracy tuple and authoritative bottleneck order.
V16.ACC.2	Report all uncertainty axes separately before optional combination.
V16.VAL.1	Pass source, matching, rank, evolution, threshold, nuclear, and process-readiness benchmarks.
V16.ISO.1	Keep all new objects disconnected from the accepted 216-route production model until a separately authorized migration.

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